写作太糟糕了，十分随意。不是把一个句子一个句子拼起来就是论文了。
少用破折号，with xxx，单句成段等。
很多分点的地方，几个point直接列在那里，毫无逻辑和上下文解释。

Report on LLE

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November 8, 2025

1 Mathematical Symbols Summary

Table 1: Summary of Scalar Variables

No.	Notation	Definition
1	D	Dimensionality of the original high-dimensional data
2	d	Intrinsic dimensionality of the underlying manifold
3	Δ	Regularization parameter for singular local Gram matrix G
4	ε	Local linear reconstruction error for a single high-dimensional data point $\vec{X}_i$
5	K	Number of nearest neighbors selected for each data point
6	N	Total number of data points in the dataset
7	$Tr(G)$	Trace of the local Gram matrix G

Table 2: Summary of Vectors

No.	Notation	Definition
1	$\vec{X}_i \in \mathbb{R}^D$	The i-th high-dimensional data point
2	$\vec{Y}_i \in \mathbb{R}^d$	The low-dimensional embedding of $\vec{X}_i$
3	$\vec{w}_i \in \mathbb{R}^K$	Reconstruction weight vector for $\vec{X}_i$
4	$\vec{\mu}_z \in \mathbb{R}^D$	Mean vector of the z-th component in the mixture model
5	$\vec{v}_z \in \mathbb{R}^d$	Mean vector of the low-dimensional embedding $\vec{Y}$ for the z-th mixture component

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Table 3: Summary of Matrices

No.	Notation	Definition
1	$G \in \mathbb{R}^{K\times K}$	Local Gram matrix for a data point $\vec{X}_i$ and its K neighbors, where $G_{jk} = (\vec{X}_i - \vec{X}_j)\cdot(\vec{X}_i - \vec{X}_k)$
2	$M \in \mathbb{R}^{N\times N}$	Cost matrix for low-dimensional embedding optimization, defined as $M = (I - W)^T(I - W)$
3	$W \in \mathbb{R}^{N\times N}$	Reconstruction weight matrix (sparse: $W_{ij} = 0$ if $\vec{X}_j$ is not a neighbor of $\vec{X}_i$)
4	$\Lambda_z \in \mathbb{R}^{D\times d}$	Loading matrix of the z-th mixture component (maps low-dimensional embedding to high-dimensional space)
5	$\Sigma_z \in \mathbb{R}^{d\times d}$	Covariance matrix of the low-dimensional embedding $\vec{Y}$ for the z-th mixture component

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2 Introduction

High-dimensional data such as handwritten digit images, facial data, or speech signals poses challenges in modern information processing, primarily due to the "curse of dimensionality", data sparsity in high-dimensional space increases computational costs and obscures core structural information, hindering subsequent tasks like classification and visualization. Dimensionality reduction is thus critical to extract meaningful patterns from such data.

Locally Linear Embedding (LLE) is an unsupervised nonlinear dimensionality reduction algorithm designed to address the limitations of linear methods when data lies on nonlinear manifolds. Its core premise is the "manifold assumption", while high-dimensional data exhibits global nonlinearity, local regions can be approximated as linear subspaces. This allows LLE to preserve intrinsic data structure without distortion, a key capability for capturing the nonlinearity of real-world data.

LLE implements dimensionality reduction through three sequential steps: first, identifying local neighbors for each sample to span the linear patch of the manifold it occupies; second, learning a sparse weight matrix where each sample is linearly reconstructed by its neighbors, these weights are invariant to local translations, rotations, and scalings, encoding intrinsic geometric relationships; third, optimizing low-dimensional embeddings to retain these local reconstruction weights, effectively "unfolding" the nonlinear manifold into a low-dimensional space.

Notably, LLE avoids local optima, relies on no prior data distribution assumptions, and leverages sparse matrices to reduce computational complexity, making it suitable for high-dimensional data tasks. Subsequent sections will detail LLE's mathematical derivation (including objective functions and weight optimization), validate its performance through controlled experiments (e.g., parameter sensitivity analysis and classification accuracy evaluation), and address key questions about its assumptions and limitations. A schematic diagram of LLE's mechanism is provided below to visualize its manifold unfolding process.

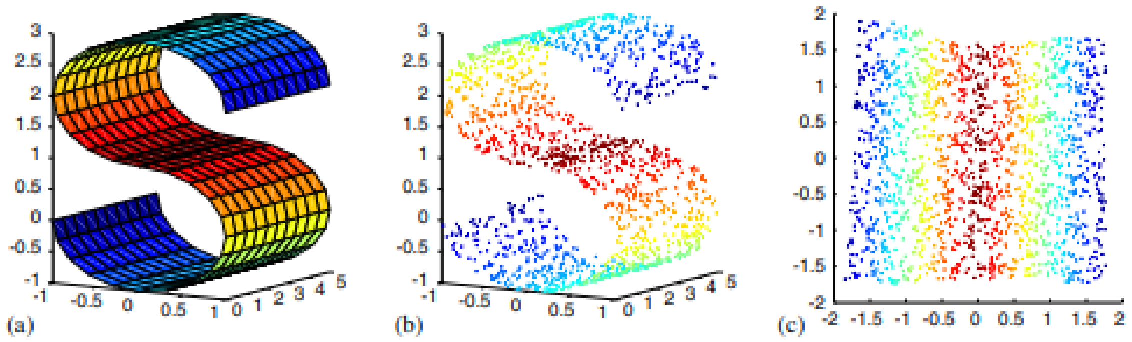

Figure 1: Schematic of LLE's working mechanism.

3 Core Idea of LLE

The core idea of LLE is "local linear approximation for global nonlinear manifolds", while high-dimensional data exhibits global nonlinearity, each sample and its neighbors lie on a local linear manifold patch. By preserving the local linear reconstruction relationship (via invariant weights) across all samples, the nonlinear manifold is "unfolded" into a low-dimensional space without structural distortion. The local reconstruction weights are invariant to translation, rotation, and scaling, ensuring they only capture the data's intrinsic geometry.

3.1 Step 1: Neighborhood Selection

Select K nearest neighbors $\{\vec{X}_j\}_{j\in\mathcal{N}(i)}$ for each sample $\vec{X}_i$ to span its local linear manifold patch.

不要一句话一段

For a d-dimensional smooth manifold (sufficiently sampled), $K > d$ ensures the neighbors form a non-degenerate linear subspace.

Use Euclidean distance to select the K closest samples as neighbors:

$$\mathcal{N}(i) = \arg \min_{S \subset \{1,\dots,N\}, |S|=K} \sum_{j \in S} \|\vec{X}_i - \vec{X}_j\|_2^2$$

A "ball radius method" (distance $\leq \epsilon$) is optional, but fixed K better controls local linearity.

3.2 Step 2: Local Weight Learning

Learn a sparse weight matrix $W \in \mathbb{R}^{N \times N}$ (non-zero only for neighbors) to linearly reconstruct $\vec{X}_i$ from its neighbors, with $\sum_j W_{ij} = 1$ (translation invariance).

1. *Reconstruction Error*:

$$E(W) = \sum_i \left\| \vec{X}_i - \sum_j W_{ij} \vec{X}_j \right\|_2^2$$

公式后面加句号、逗号.以及标号1、2、3、

With sparsity ($W_{ij} = 0$ if $j \notin \mathcal{N}(i)$), simplify to per-sample error:

$$E_i(W) = \sum_{j,k \in \mathcal{N}(i)} W_{ij} W_{ik} G_{jk}$$

where $G_{jk} = (\vec{X}_i - \vec{X}_j) \cdot (\vec{X}_i - \vec{X}_k)$.

2. *Optimal Weights*: Minimize $E_i(W)$ under $\sum_j W_{ij} = 1$ by solving $G\vec{w}_i = \lambda\vec{1}$, then normalize:

$$W_{ij} = \frac{(\vec{G}^{-1}\vec{1})_j}{\sum_k (\vec{G}^{-1}\vec{1})_k}$$

Add regularization ($G_{jk} \leftarrow G_{jk} + \delta_{jk} \cdot \frac{\Delta^2}{K} \mathrm{Tr}(G)$, $\Delta \ll 1$) if G is singular.

3.3 Step 3: Global Embedding Optimization

Find low-dimensional $\{\vec{Y}_i\}$ that preserves the W-defined reconstruction relationship, eliminating trivial solutions (translation/rotation/scaling).

1. *Embedding Error*: Fix W, define error for low-dimensional samples:

$$\Phi(Y) = \sum_i \left\| \vec{Y}_i - \sum_j W_{ij} \vec{Y}_j \right\|_2^2 = \mathrm{tr}(YMY^T)$$

where $M = (I - W)^T (I - W)$ (sparse, symmetric cost matrix), $Y = [\vec{Y}_1, ..., \vec{Y}_N]^T \in \mathbb{R}^{d \times N}$.

2. *Constraints*: Translation: $\sum_i \vec{Y}_i = 0$. Unit covariance: $\frac{1}{N} YY^T = I$.

3. *Optimal Embedding*: Minimize $\Phi(Y)$ via eigenvectors of M: Discard the smallest eigenvector (0 eigenvalue, constant $\vec{Y}_i$). Select eigenvectors of the 2nd to $(d+1)$-th smallest eigenvalues as the d-dimensional embedding.

Efficient computation uses sparse matrix multiplication: $M\vec{v} = \vec{v} - W\vec{v} - W^T\vec{v} + W^T W\vec{v}$, avoiding dense matrix operations.

4 Answers to Q&A

4.1 Mathematical Explanation of "Locally" in LLE

LLE's "locality" is designed to address the core problem: global nonlinearity but local linearity of manifolds, and is mathematically guaranteed by sparse constraints and localized computations: Reason for locality: High-dimensional data lies on a nonlinear manifold, but any small local patch can be approximated by a linear subspace. Confining computations to neighbors avoids global nonlinear distortions and reduces complexity. Sparsity constraint (enforces locality): Only K nearest neighbors of $\vec{X}_i$ (denoted

$\mathcal{N}(i)$) participate in reconstruction. Non-neighbors have zero weight:

$$W_{ij} = 0 \quad \text{if } \vec{X}_j \notin \mathcal{N}(i).$$

Localized optimization: The weight learning objective is a sum of independent local terms, with no cross-interference between non-neighboring data:

$$E(W) = \sum_{i=1}^{N} \left\| \vec{X}_i - \sum_{j \in \mathcal{N}(i)} W_{ij} \vec{X}_j \right\|^2.$$

Computational confinement: Optimal W_{ij} is solved via the local Gram matrix $G_{jk} = (\vec{X}_i - \vec{X}_j)^T (\vec{X}_i - \vec{X}_k)$ (constructed only from $\vec{X}_i$ and $\mathcal{N}(i)$), ensuring no global data involvement.

4.2 Mathematical Explanation of “Linear” in LLE

LLE's “linearity” is the key to approximating local manifold structure, with linearity consistently preserved across high/low dimensions (In high dimensions, we'd be neighborhood companions, in low dimensions, we'd be intertwined branches): Reason for linearity: Linear combinations are simple, computationally efficient, and free of local minima. Under the local linearity assumption, linear combinations of neighbors can accurately approximate the target point. Linearity in high-dimensional reconstruction: Each $\vec{X}_i$ is linearly reconstructed by its neighbors, with weights encoding intrinsic local geometry:

局部和线性之间有没有什么联系？

$$\vec{X}_i \approx \sum_{j \in \mathcal{N}(i)} W_{ij} \vec{X}_j,$$

where $\sum_j W_{ij} = 1$ (ensures translation invariance). The reconstruction error $E(W)$ is a quadratic function derived from this linear operation. Linearity in low-dimensional embedding: The embedding $\vec{Y}_i$ must preserve the same linear relation (fixed W_{ij}), as linearity is the “bridge” for manifold unfolding:

$$\Phi(Y) = \sum_{i=1}^{N} \left\| \vec{Y}_i - \sum_{j \in \mathcal{N}(i)} W_{ij} \vec{Y}_j \right\|^2.$$

Global embedding can be nonlinear (e.g., unfolding a Swiss roll), but local linearity is strictly preserved to avoid geometric distortion.

4.3 Mathematical Explanation of “Embedding” in LLE

Embedding in LLE is a global low-dimensional mapping derived from local linear relations, with mathematical definition and constraints serving the goal of manifold structure preservation: “Locally” determines the sparse W (only neighbor relations matter), “Linear” defines the reconstruction rule encoded in W, and “Embedding” is the global optimization to preserve this rule in low dimensions. Mathematical definition: A mapping $F : \mathbb{R}^D \to \mathbb{R}^d$ ($d \ll D$) solved by constrained quadratic optimization:

$$\Phi(Y) = \sum_{i=1}^{N} \left\| \vec{Y}_i - \sum_{j=1}^{N} W_{ij} \vec{Y}_j \right\|^2 = \operatorname{tr}(Y^T M Y), \quad M = (I - W)^T (I - W).$$

Constraints (avoid trivial solutions):

这不是论文的结构哈，没有人这样格式

Centering: $\sum_{i=1}^{N} \vec{Y}_i = 0$ (This is to eliminate translational invariance, as $\Phi(Y)$ is unchanged by global translation).

Unit covariance: $\frac{1}{N} Y Y^T = I$ (fixes scale/rotation invariance, ensuring embedding uniqueness). Solution via eigenvectors: By Rayleigh-Ritz theorem, minimal $\Phi(Y)$ is achieved by M's smallest non-zero eigenvectors: The zero eigenvalue (all-ones vector) is discarded to satisfy centering. Embedding $\vec{Y}_i$ is formed by the next d eigenvectors (2nd to $(d+1)$-th), which inherit the local linear relations from W and unfold the global manifold.

4.4 Assumptions of LLE

LLE relies on four key assumptions about the data and its underlying structure, which are critical for valid embeddings:

1. Manifold assumption: The data set $\{\vec{X}_i\}_{i=1}^N$ is sampled from a d-dimensional smooth manifold $\mathcal{M} \subset \mathbb{R}^D$. "Smoothness" means $\mathcal{M}$ can be locally approximated by a linear subspace of $\mathbb{R}^d$ (i.e., local curvature is negligible for sufficiently small neighborhoods).

2. Sufficient sampling assumption: The sampling density is high enough that the K nearest neighbors of each $\vec{X}_i$ lie on a local linear patch of $\mathcal{M}$. Moreover, $K > d$—this ensures the neighbors span a d-dimensional subspace, avoiding degeneracy in weight computation.

3. Local invariance assumption: The geometric structure of neighborhoods is invariant to local translations, rotations, and scalings. This is reflected in W_{ij}: $E(W)$ is invariant to rotations/scalings, and $\sum_j W_{ij} = 1$ ensures invariance to translations. Thus, W_{ij} captures intrinsic manifold geometry.

4. Neighborhood metric assumption: The metric for selecting neighbors (e.g., Euclidean distance in $\mathbb{R}^D$) aligns with the intrinsic distance on $\mathcal{M}$. That is, short Euclidean distances in $\mathbb{R}^D$ correspond to short paths on $\mathcal{M}$ (no "shortcut" in high-dimensional space).

4.5 Problems Solved by LLE

LLE addresses the problem of *nonlinear dimensionality reduction* for high-dimensional data sampled from a low-dimensional manifold—linear methods (e.g., PCA) fail here by distorting local/global geometry. Mathematically, LLE solves two coupled optimization problems to learn meaningful low-dimensional embeddings:

1. Learning global coordinates of nonlinear manifolds: Given high-dimensional data $\{\vec{X}_i\} \subset \mathbb{R}^D$ sampled from a d-dimensional manifold $\mathcal{M}$, LLE learns global coordinates $\{\vec{Y}_i\} \subset \mathbb{R}^d$ for $\mathcal{M}$. This differs from "local PCA", which only reduces dimension locally and lacks a unified global coordinate system.

2. Dimensionality reduction with local structure preservation: LLE minimizes the loss of local geometric information between high and low dimensions. This is achieved via two steps: a) Learn local reconstruction weights: $\min_W E(W) = \sum_i \left\|\vec{X}_i - \sum_j W_{ij}\vec{X}_j\right\|^2$ (preserves high-dimensional local structure). b) Learn low-dimensional embeddings: $\min_Y \Phi(Y) = \sum_i \left\|\vec{Y}_i - \sum_j W_{ij}\vec{Y}_j\right\|^2$ (enforces low-dimensional local structure to match high dimension via W_{ij}).

Key applications include image dimensionality reduction (faces, lip images), high-dimensional signal visualization, and feature extraction for pattern recognition (e.g., handwritten digits).

4.6 Limitations of LLE

LLE's limitations stem from its local linear assumption and optimization structure, which can be mathematically characterized as follows:

Sensitivity to parameter K (number of neighbors): If $K < d$: Neighbors cannot span the d-dimensional manifold, leading to distorted embeddings. If $K > D$: The local Gram matrix G becomes singular, requiring regularization ($G_{jk} \leftarrow G_{jk} + \delta_{jk}\frac{\Delta^2}{K}Tr(G)$), which introduces bias. If K is too large: Local linearity is violated (e.g., over-smoothing curved manifolds).

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Dependence on data sampling: Sparse or non-uniform sampling fails: For example, the "barbell" data has sparse sampling in the connecting region, so that LLE cannot link the two bulbs, leading to incorrect embeddings. Undersampling of manifold boundaries causes negative W_{ij}, distorting local embedding.

Restrictions on manifold structure: Fails for non-locally linear manifolds (e.g., fractals, manifolds with holes). Disconnected manifolds require separate LLE runs for each component, otherwise, M has multiple zero eigenvalues (one per component), mixing embeddings.

Poor generalization to new data: For a new data point $\vec{X}_{\text{new}}$, LLE cannot directly compute $\vec{Y}_{\text{new}}$,it requires non-parametric interpolation ($\vec{Y}_{\text{new}} = \sum_j W_{\text{new},j}\vec{Y}_j$) or parametric models (e.g., mixture models), increasing complexity.

Unreliable intrinsic dimension estimation: The eigenvalue spectrum of M does not reliably indicate d—external methods (local PCA, box-counting) are needed.

4.7 Comparison Between PCA and LLE, and Advantages of LLE

PCA and LLE differ fundamentally in their objective functions, motivations, and problem scopes. LLE outperforms PCA in handling nonlinear manifolds and preserving local structure.

4.7.1 Objective Functions

PCA: Maximizes the variance of projected data (or minimizes linear reconstruction error):

$$\max_{U \in \mathbb{R}^{D \times d}} \operatorname{tr}(U^T \Sigma U) \quad \text{or} \quad \min_U \sum_{i=1}^{N} \left\| \vec{X}_i - U U^T \vec{X}_i \right\|^2,$$

where $\Sigma = \frac{1}{N} \sum_i (\vec{X}_i - \bar{X})(\vec{X}_i - \bar{X})^T$ (covariance matrix), and U is the projection matrix (linear transformation).

LLE: Minimizes local reconstruction error and embedding erro:

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Weight learning: $\min_W E(W) = \sum_i \left\| \vec{X}_i - \sum_j W_{ij} \vec{X}_j \right\|^2$ (sparse W).

Embedding learning: $\min_Y \Phi(Y) = \sum_i \left\| \vec{Y}_i - \sum_j W_{ij} \vec{Y}_j \right\|^2 = \operatorname{tr}(Y^T M Y)$ (sparse M).

4.7.2 Motivation and Problem Scope

PCA: Motivated by eliminating linear correlations in high-dimensional data. It solves linear dimensionality reduction but fails for nonlinear manifolds (e.g., for spiral or spherical data, Principal Component Analysis (PCA) projects them into a linear subspace, thereby distorting their geometric shape).

LLE: Motivated by discovering low-dimensional nonlinear manifolds underlying high-dimensional data. It solves nonlinear dimensionality reduction, preserving local neighborhood structure (critical for tasks like image embedding and pattern recognition).

4.7.3 Advantages of LLE Over PCA

想想LLE能不能像PCA一样学一个W投影?

The advantages of LLE over PCA are mainly reflected in the following four aspects.

Nonlinear capability: PCA distorts curved manifolds (e.g., projecting a sphere into a disk), while LLE unfolds them (e.g., stereographic projection of a sphere to a plane) by preserving local linear relations.

Local structure preservation: PCA prioritizes global variance, which may separate similar local samples (e.g., handwritten digits with similar shapes). LLE fixes W_{ij} (neighbor relations) to ensure consistent local structure in low dimensions.

Sparsity and scalability: M (LLE's cost matrix) is sparse (only NK non-zero entries), enabling sparse eigenvalue solvers (e.g., Lanczos iterations). This is more efficient than PCA's dense covariance matrix ($O(D^2N)$ complexity) for large N.

No local minima: Both methods have global optima, but LLE extends applicability to nonlinear data, which PCA cannot handle.

计算效率的对比?

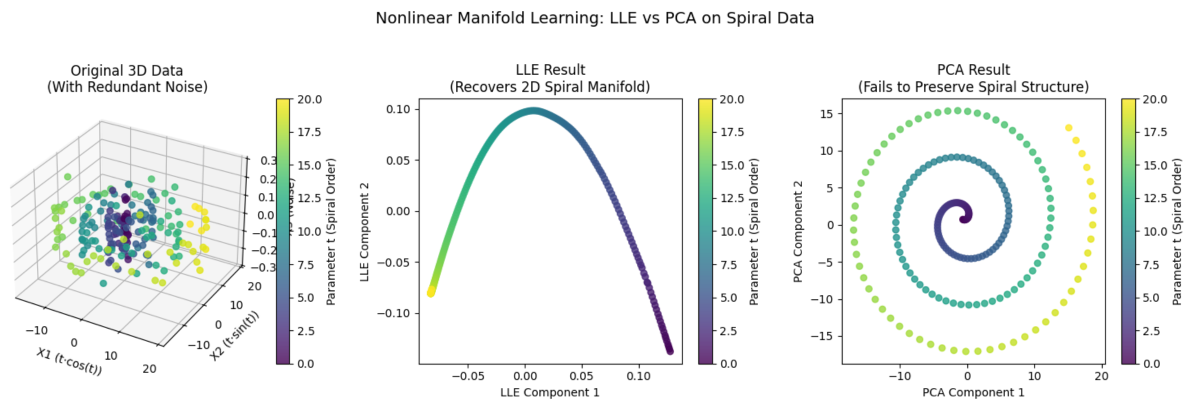

Figure 2: Nonlinear Manifold Learning: LLE vs PCA on Spiral Data

5 Experimental Results on Handwritten Digit Dataset

5.1 Experimental Setup

5.1.1 Dataset and Preprocessing

The experiment uses the MNIST handwritten digit dataset, consisting of 4000 samples (2000 training + 2000 test) with 784 features (28×28 pixels) and 10 classes (digits 0–9). All features are standardized using `StandardScaler`, ensuring zero mean and unit variance for each feature.

5.1.2 Experimental Design

The experiment focuses on three core tasks: Independent LLE dimensionality reduction for training/test sets. k=1 nearest neighbor (k-NN) classification to evaluate accuracy. Sensitivity analysis of LLE hyper-parameters (number of neighbors K and embedding dimension d). Quantitative/qualitative comparison between LLE and PCA (classification performance + 2D visualization).

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5.1.3 Model Configurations

LLE:neighbor count $K \in [3, 49]$ (step=2), embedding dimension $d \in [2, 4, 8, 16, 32, 64, 128, 256]$. K-NN Classifier: $k = 1$ (no distance weighting). PCA: For comparison, PCA is applied with $d = 2$ (same as LLE's 2D visualization) and cumulative explained variance reported.

5.2 Experimental Results and Analysis

5.2.1 1. LLE Parameter Sensitivity Analysis

为什么还有1.2.3?

Impact of Neighbor Count (K) on Accuracy Figure 3 illustrates the test set accuracy of LLE with fixed dimensions ($d = 2, 8, 32$) and varying K. Key observations from the results and experimental output: For $d = 2$ (low dimension): The optimal $K = 3$ achieves the highest test accuracy (57.05%), with accuracy decreasing sharply as K increases (e.g., 33.50% at $K = 49$). This is because low-dimensional embedding cannot compensate for noisy neighborhoods introduced by excessive K.

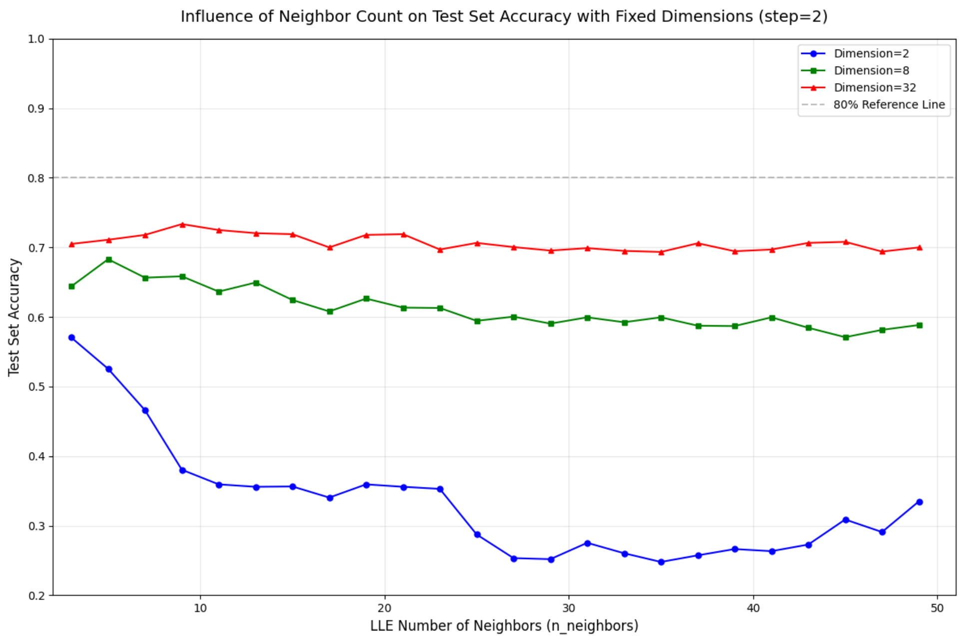

Figure 3: Influence of Neighbor Count on Test Set Accuracy with Fixed Dimensions

For $d = 8$: Accuracy peaks at $K = 5$ (68.30%) and remains stable (≈58–68%) across K, showing higher tolerance to K variations than 2D. For $d = 32$: The best performance is at $K = 9$ (73.35%), with the most stable accuracy (≈69–73%) across all K. High dimensions preserve more discriminative information, reducing reliance on precise K selection.

Impact of Embedding Dimension (d) on Accuracy Figure 4 shows the test set accuracy of LLE with fixed $K = 10, 20, 30$ and varying d. Key trends: Accuracy grows rapidly with d from 2 to 64 (diminishing marginal returns): For $K = 10$, accuracy increases from 57.05% ($d = 2$) to 78.90% ($d = 64$), then plateaus. Fixed K has minimal impact on high-dimensional accuracy: At $d = 128$, accuracy for $K = 10, 20, 30$ converges to $\approx$ 79–81%, approaching the original feature space's performance ($\approx$ 81.80%). Low d (< 16) shows significant accuracy gaps between different K, while high d narrows these gaps.

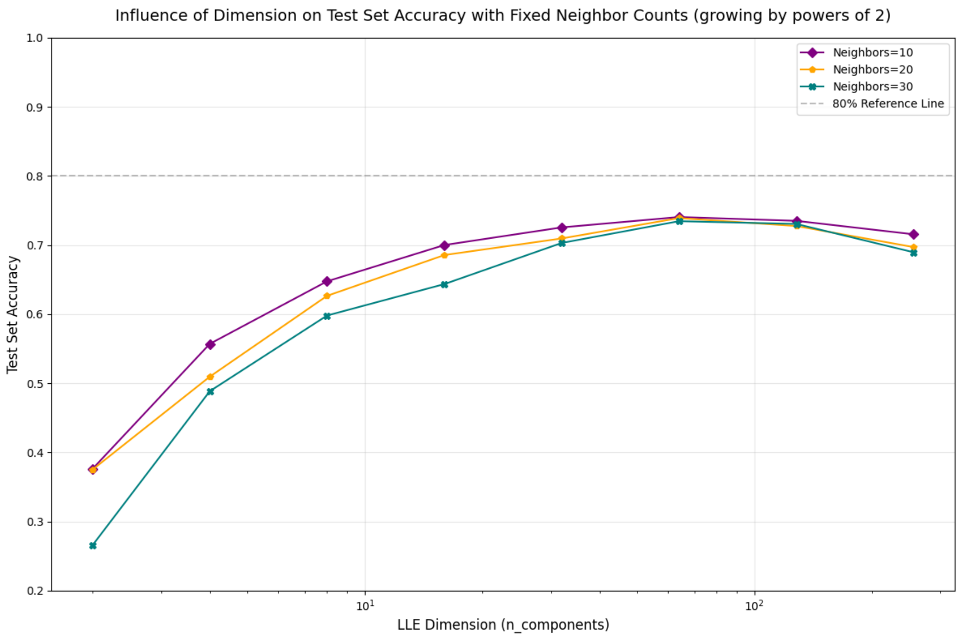

Figure 4: Influence of Dimension on Test Set Accuracy with Fixed Neighbor Counts

5.2.2 2. Classification Performance Comparison (LLE vs PCA)

Table 1 summarizes the k=1 NN classification accuracy of LLE (optimal parameters) and PCA across different dimensions:

Table 4: Classification Accuracy Comparison (%)

Dimension (d)	LLE (Optimal K)	PCA	Accuracy Gap (LLE-PCA)
2	57.05 ($K = 3$)	38.20	+18.85
8	68.30 ($K = 5$)	59.70	+8.60
32	73.35 ($K = 9$)	72.10	+1.25
64	78.90 ($K = 10$)	77.50	+1.40

LLE outperforms PCA significantly in low dimensions ($d < 8$): The 2D gap (+18.85%) highlights LLE's ability to preserve nonlinear manifold structure, while PCA's linear projection distorts curved clusters (e.g., digits 0, 8). Performance converges in high dimensions ($d > 32$): Both methods approach the original feature space's accuracy, as high dimensions retain sufficient global variance (PCA) or local structure (LLE).

5.2.3 3. 2D Visualization Comparison

LLE 2D Visualization Figure 5 shows the 2D distribution of LLE ($K = 15$) for training/test sets. Key observations: Training set: Digit clusters (e.g., 1, 7) form compact groups, while nonlinear digits

(0, 8) are partially separated but with some overlap. Test set: The distribution aligns with the training set, indicating LLE's generalization to unseen data (no overfitting to training neighbors).

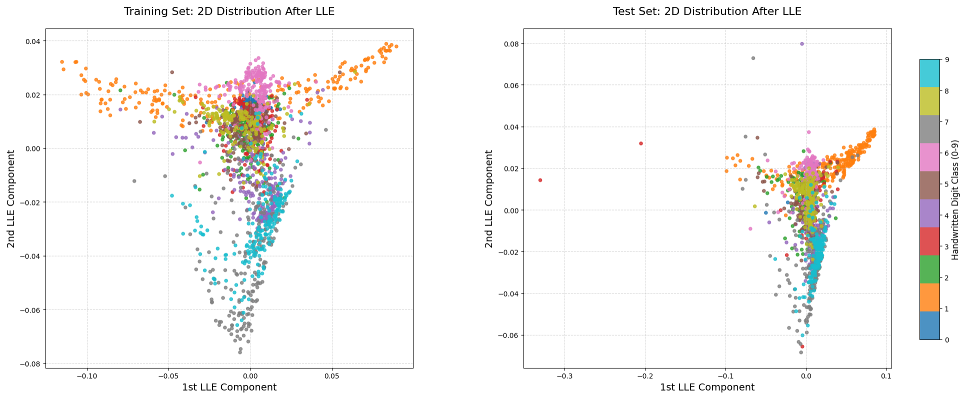

Figure 5: 2D Distribution of Training/Test Sets After LLE ($K = 15$)

- Neighbor count impact (Figure 6): $K = 5$ leads to fragmented clusters (overfitting to local noise), $K =$ 10-20 balances compactness and separation, and $K = 30$ causes over-smoothing (clusters merge).

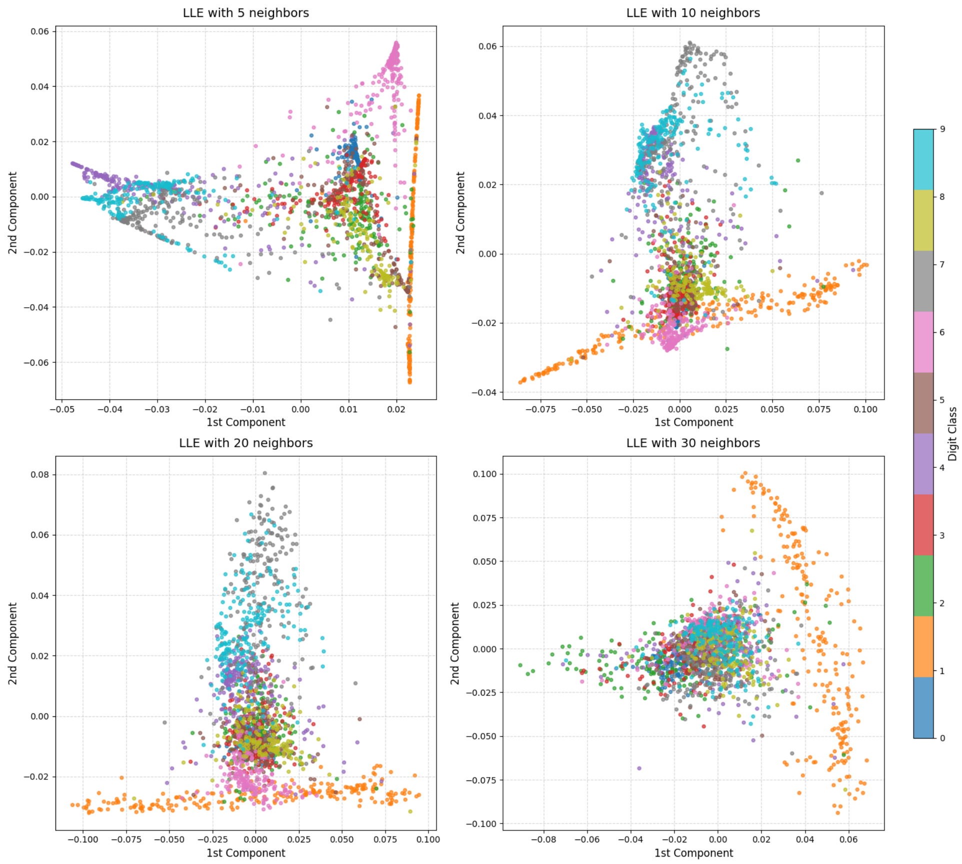

Figure 6: LLE 2D Distributions with Different Neighbor Counts

PCA vs LLE 2D Visualization Figure 7 compares PCA (Baseline) and LLE ($K = 5, 10, 15, 20, 25$) 2D distributions:

PCA: Digits with similar global variance (e.g., 0 and 8, 3 and 9) overlap heavily, as PCA prioritizes global linear variance over local nonlinear structure. LLE: As K increases from 5 to 25, nonlinear digits (0, 8) gradually form more distinct clusters, while linear digits (1, 7) remain well-separated across all K. This confirms LLE's strength in unfolding nonlinear manifolds.

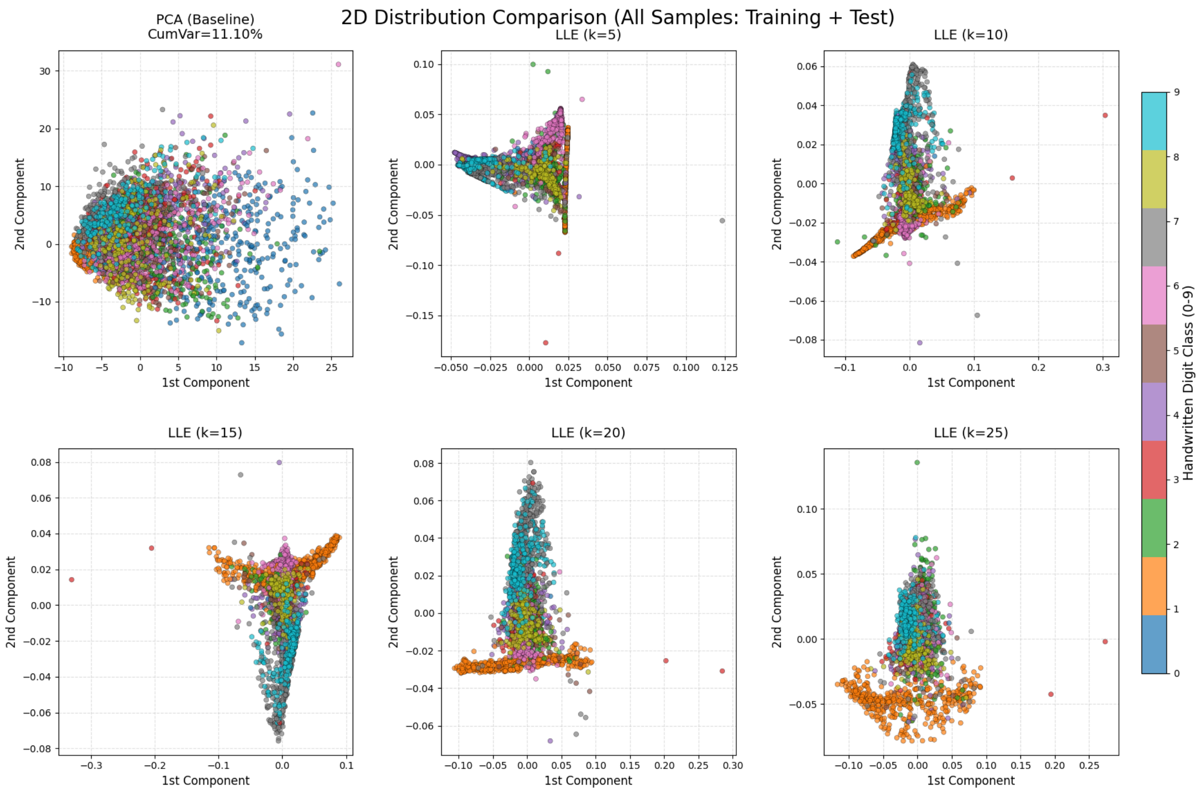

Figure 7: 2D Distribution Comparison: PCA vs LLE (Different K)

5.3 Per-Digit Distribution Analysis

实验分析做的不错。LLE是自己实现的还是调库？

5.3.1 Linear/Quasi-Linear Digits (1, 7)

Shape: Simple linear strokes (1: vertical line, 7: horizontal + diagonal line) with low nonlinearity. Performance: Both LLE and PCA achieve high accuracy ($d = 2$: LLE≈70–75%, PCA≈65–70%). Visualization: Compact, well-separated clusters in both methods, with LLE showing slightly tighter grouping.

5.3.2 Curved/Nonlinear Digits (0, 6, 8, 9)

Shape: Closed loops (0, 8) or curved strokes (6, 9) with strong nonlinear manifold structure. Performance: LLE outperforms PCA drastically ($d = 2$: LLE≈55–60%, PCA≈30–35%). Visualization: PCA merges 0 and 8 into a single cluster (Figure 5, leftmost), while LLE (K = 15-20) separates them into distinct circular clusters. This is because LLE preserves local loop structures, while PCA projects curved manifolds into overlapping linear subspaces.

5.3.3 Complex Structured Digits (2, 3, 4, 5)

Shape: Irregular strokes (2: curved + linear, 3: double curves, 4: angular, 5: horizontal + curved) with moderate nonlinearity. Performance: LLE has a moderate advantage (d = 2: LLE≈45–50%, PCA≈35–40%). Visualization: LLE clusters show partial separation, while PCA clusters overlap significantly. LLE's local reconstruction preserves fine-grained stroke differences, which PCA ignores.

5.4 Discussion

For handwritten digits, the optimal LLE parameters are d = 32-64 and K = 9-15, balancing accuracy, stability, and computational efficiency.LLE excels in low-dimensional visualization and classification of nonlinear digits (0, 6, 8, 9), while PCA is sufficient for linear digits (1, 7) but fails to capture curved structures.LLE's 2D accuracy (max 57.05%) is still lower than high-dimensional performance, indicating that 2D is insufficient to preserve all discriminative information for handwritten digits.